R18

Code No: 152AG

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year II Semester (Special) Examinations, January - 2021
ENGINEERING GRAPHICS
(Computer Science and Engineering)

Time: 2 hours **Max. Marks: 75**

Answer any three questions
All questions carry equal marks
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1.a) Draw a vernier scale of R.F. = 5 to read 1/5 cm and 1/25 cm and to measure up to 5 cm. Mark on the scale distances of 2.12 cm and 4.29 cm.

b) The major axis of an ellipse is 150 mm long and the minor axis is 100 mm long. Find the foci and draw the ellipse. Draw a tangent to the ellipse at a point on it 25 mm above the major axis. [10+15]

2. A hexagonal plate of side 20 mm rests in the HP on one of its sides inclined at 45^0 to the VP. The surface of the plate makes an angle of 30^0 with the HP. Draw the front and top views of the plate. [25]

3. The point A of a line AB is in HP and 60 mm in front of VP. The point B is in VP and 40 mm above HP. The distance between projectors is 70 mm. Draw the projections of the line and find the true length and inclinations. [25]

4. A pentagonal pyramid, having side of base 30 mm and axis 90 mm long, is resting on its base with one of its base edges parallel, nearer, and 15 mm away from VP. It is cut by a plane perpendicular to HP, inclined at 40^0 to VP, and 10 mm away from the axis. Draw the sectional top view and front view. Also draw the true shape of the section. [25]

5. A lamp shade is formed by cutting a cone of base 144 mm diameter and 174 mm height by a horizontal plane at a distance of 72 mm from the apex and by another plane inclined at 30^0 to HP and passing through one extremity of the base. Draw the development of the lamp shade. [25]

6. A cube of 30 mm side rests centrally on the top of a cylindrical slab of 50 mm diameter base and 40 mm height lying on the ground. The axis of both solids are in the same straight line, and the base sides of the cube are equally inclined to the V.P. Draw the isometric projection of the solid. [25]

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